

Subcontracting Specification, BOM Overview and Manufacturing Strategy

Drone Transport Project

15 de mayo de 2026

1. Clarification of the subcontracting work

The subcontracting work was carried out by Maelle Kouidri group. However, their contribution is limited only to the **propeller drawings**. It is important to clearly define the scope of the subcontracting: the subcontractor did not design the complete drone, did not manufacture any part, did not create the final assembly and did not validate the full BOM.

Their real contribution was the geometric documentation of the propellers. The drawing provided by the subcontractor defines the main dimensions used as references for the CAD model. In particular, it includes:

- a two-blade propeller with an outside diameter of 383 mm;
- a three-blade propeller with an outside diameter of 338.7 mm;
- a maximum thickness of 20 mm;
- the hub centre, used as the reference for mounting the propeller on the motor shaft.

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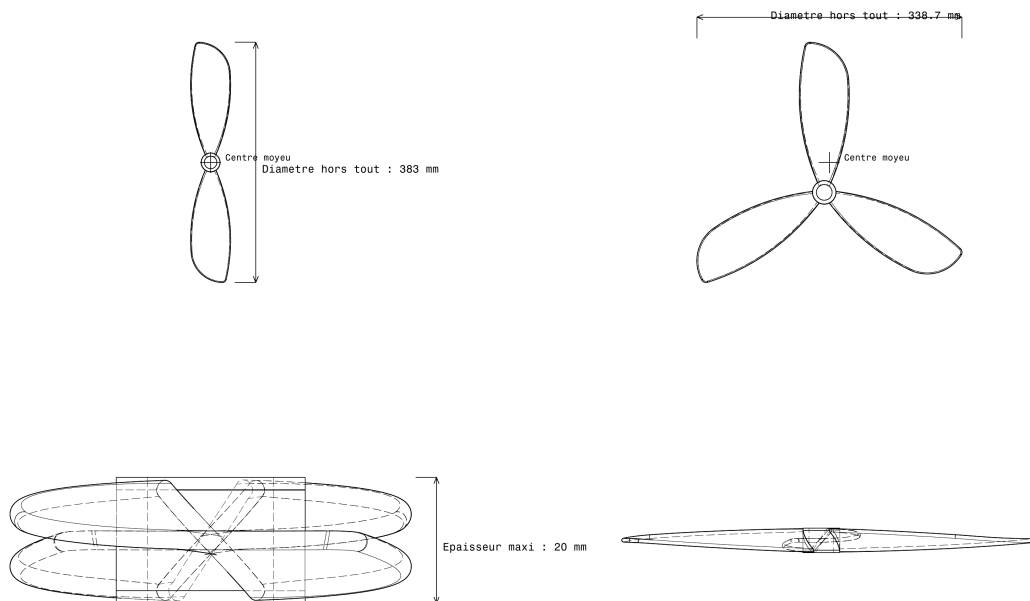


Figura 1: Propeller drawing produced by the subcontractor: main dimensions, outside diameters, maximum thickness and hub centre.

The propeller was not physically manufactured by the subcontractor. This is correct because the propeller is a critical flight component. During operation, it rotates at high speed, generates thrust, transmits vibrations to the motor and is subjected to aerodynamic loads. Therefore, it cannot be accepted as a flight-ready component without dimensional control, dynamic balancing and material validation.

The expected result was not a manufactured propeller, but a **parametric** and **machinable** propeller model. Parametric means that the diameter, number of blades, thickness and hub geometry can be modified in the CAD model. Machinable means that the geometry can be transformed into a real manufacturable part through a controlled process such as CNC machining, a machined mould, composite manufacturing or supplier production.



Figura 2: Render of the modelled propeller. The render validates the visual shape, but it does not validate the part for flight.

2. Work completed and work not completed by the subcontractor

2.1. Work completed

The subcontractor completed the following work:

- production of the propeller drawing;
- definition of the main propeller dimensions;
- identification of the hub centre;
- representation of a two-blade and a three-blade propeller;
- delivery of a dimensional reference useful for CAD modelling.

2.2. Work not completed

The subcontractor did not complete the following tasks:

- no physical manufacturing of the propeller;
- no flight validation of the propeller;
- no dynamic balancing;
- no manufacturing of drone parts;
- no final assembly of the drone;
- no complete material definition for all parts;
- no full BOM preparation;
- no interference validation in CATIA.

This clarification avoids attributing work to the subcontractor that was not actually done. The propeller drawing is part of the subcontracting work, but the manufacturing and validation of the drone remain the responsibility of the main project group.

3. BOM provided separately in Excel

The complete BOM is not included in this LaTeX document because a full table would be too long and difficult to read. Instead, it is provided in a separate Excel file:

BOM_drone_english_dynamic_colors.xlsx

The Excel file contains the list of parts, materials and manufacturing methods. It also clearly separates parts manufactured by the team from components purchased from suppliers.

The BOM is organised to make the project easier to verify. Each line identifies the part, its function, the material used and the selected manufacturing process, such as:

- laser cutting;
- CNC milling;
- 3D printing;
- tube cutting and drilling;
- supplier purchase;
- assembly only.

This separation is more practical than placing the full BOM directly in the report, because Excel allows filtering by part family, manufacturing process, material and responsibility.

4. General manufacturing strategy

The drone is divided into several families of parts. Each family uses a different manufacturing method because each one has a different mechanical function.

4.1. Structural plates

The structural plates, such as the upper plate, lower plate and internal frame, are made by the team. These parts are flat and require good dimensional precision. The selected manufacturing methods are:

- **laser cutting** for aluminium or compatible sheet material;
- **CNC milling** for thicker or more precise structural plates.

Laser cutting is suitable for flat plates because it is fast, precise and allows repeatable contours. CNC milling is also possible when better edge quality, accurate holes or thicker material are required.

4.2. Parametric arms

The arms are key structural parts because they separate the motors from the central body and transmit the thrust loads. In the CAD model, the arms are parametric: their length can be changed between 0.30 m and 0.50 m depending on the configuration of the drone.

The recommended material is carbon fibre tube because it has a high stiffness-to-weight ratio. The manufacturing process consists of:

- cutting the tube to the selected length;
- drilling the fixing holes with a jig;
- checking that all arms have the same length;
- checking the alignment between the motor axis and the drone structure.

The arms should not be made as simple 3D printed parts for flight because they are subjected to bending and vibration.

4.3. Connectors and supports

The connectors, battery support, landing gear connectors and small mechanical supports can be manufactured by:

- **3D printing** for prototypes and complex shapes;
- **CNC milling** for stronger final parts;
- **laser cutting** if the part is flat.

3D printing is useful for fast iterations, especially when checking the assembly. CNC milling is preferred when the part must resist higher loads or when better dimensional accuracy is required.

4.4. Sheet metal protectors

The sheet metal protectors are made from thin aluminium sheet. The selected process is:

- laser cutting of the flat pattern;
- bending or forming;
- deburring;
- checking the clearance with the propellers.

This process is justified because aluminium can be cut accurately by laser and provides a rigid protection with limited added mass.

4.5. Fairing and aerodynamic parts

The external fairing and aerodynamic parts have curved and complex shapes. These parts can be produced by:

- 3D printing for prototype versions;
- composite manufacturing for a final lightweight version;
- CNC machining of a mould if a composite process is selected.

The fairing is not the main structural element, but it protects the internal electronics and improves the visual and aerodynamic integration of the drone.

4.6. Propellers

The propellers are critical components. The subcontractor only produced the drawing; the propellers were not manufactured by the subcontractor.

For a prototype or visual validation, the propeller can be 3D printed. However, for flight, the propeller must be:

- purchased from a specialised supplier; or
- manufactured with a controlled composite process; or
- produced using a CNC-machined mould.

Before any flight test, the propeller must be checked for geometry, hub alignment, surface quality and dynamic balancing.

4.7. Motors

The motor assembly is represented in CAD and integrated into the drone structure. The project can design and manufacture the mechanical parts linked to the motor, such as the support, the interface with the arm and the fixing system.

However, the internal electrical motor components, such as coils, magnets, bearings and rotor elements, require specialised industrial manufacturing. For the final drone, these internal elements should be treated as supplier components unless a dedicated motor manufacturing process is available.

4.8. Electrical components and battery

Electrical components are not manufactured by the team. They are purchased and integrated into the drone. This includes:

- ESCs;
- battery;
- Pixhawk flight controller;
- Raspberry Pi;
- GPS;
- RC receiver;
- telemetry module;
- PDB;
- DC-DC power module;
- fuse module;
- camera.

The role of the team is to design the mechanical integration of these components: supports, ventilation, cable routing, accessibility and fixation.

5. Final verification required

Before manufacturing or presenting the final design, the following checks must be completed:

1. open the full assembly in CATIA;
2. check that there are no interferences between propellers, arms, protectors and fairing;
3. verify that the parametric arms correctly change between 0.30 m and 0.50 m;
4. check that the motors are correctly aligned;
5. verify that the propellers do not touch the protectors;
6. assign the correct material to each part;
7. calculate the total mass and centre of gravity;
8. export flat parts as DXF files for laser cutting;
9. export 3D printed parts as STL files;

10. export machined parts as STEP files;
11. validate the propellers through a specialised supplier or controlled manufacturing process;
12. do not perform any flight test without propeller balancing.

6. Conclusion

The subcontracting work is limited to the propeller drawing. The subcontractor did not manufacture the propeller, did not manufacture the drone parts and did not validate the full drone assembly. This clarification is important because the propeller is a critical component that requires specific validation.

The BOM is provided separately in Excel to keep the report clear and readable. The Excel file contains the full list of parts, materials and manufacturing methods. The LaTeX document explains the technical logic of the subcontracting work and the manufacturing strategy.

The final drone can be considered manufacturable if the processes are correctly separated: plates by laser cutting or CNC milling, connectors by 3D printing or CNC milling, arms by carbon tube cutting and drilling, sheet protectors by laser cutting and bending, aerodynamic parts by 3D printing or composite manufacturing, and electrical components by supplier purchase and mechanical integration.